

PROTOPULSE

# The first integrated desktop PCB prototyping station.

One machine, one software, one hour.

ProtoPulse replaces weeks of factory wait time with same-day, in-house PCB fabrication and assembly. Designed for startups, R&D labs, and universities.

~60

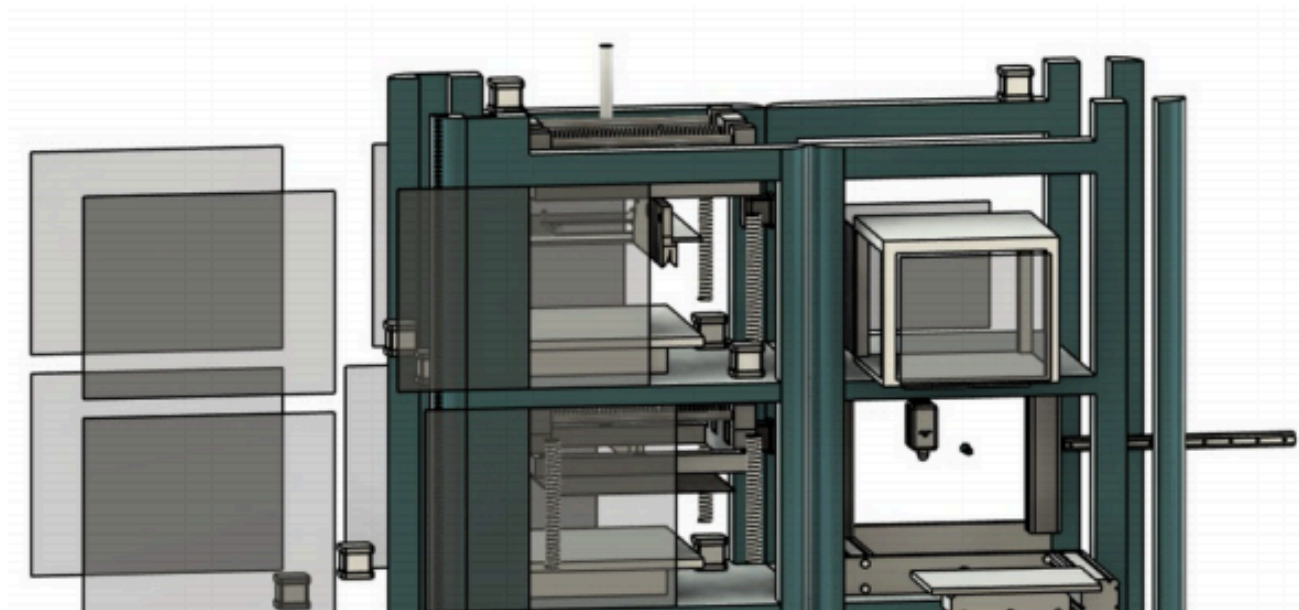
MINUTES

Gerber to assembled board

₹150

PER BOARD

Zero chemical waste



# PCB prototyping is fundamentally broken.



## Too Slow

Factories take 2-6 weeks per run. Multiple revisions stretch development to months of idle waiting time.



## Too Expensive

₹4,000-16,000+ per batch plus shipping. Minimum order quantities kill small-batch experimentation.



## Unsafe & Messy

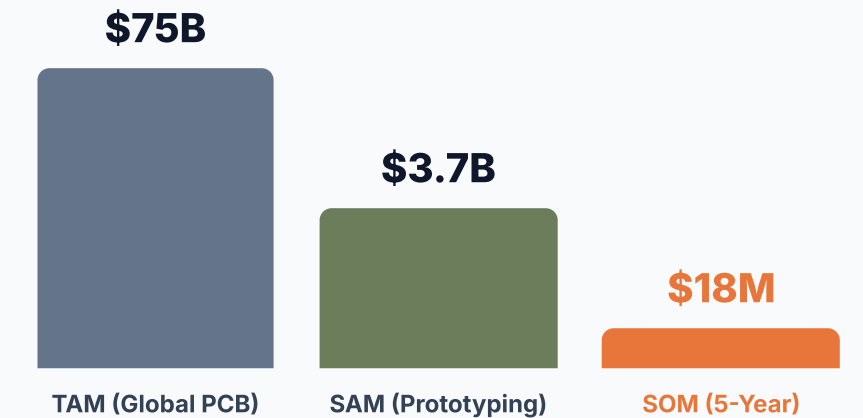
Chemical etching uses hazardous ferric chloride. Requires fume hoods, PPE, and regulated waste disposal.



## IP at Risk

Designs must be shared with external fabs. Unacceptable for defense, medical devices, and proprietary R&D.

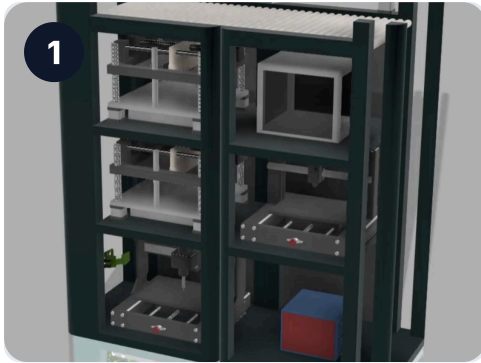
## A \$3.7B Prototyping Market



## Strategic India Opportunity

National electronics manufacturing push. 10,000+ engineering colleges. Government-funded incubators driving massive domestic demand.

# Four stages. One machine. Zero hassle.



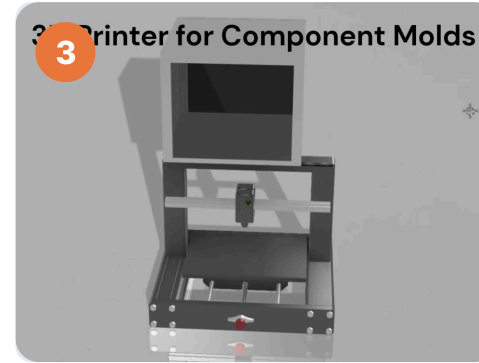
## CNC Milling

3-axis stepper system with GRBL firmware. V-bit tool carves 0.1mm isolation grooves between copper traces. Auto tool-size grouping.



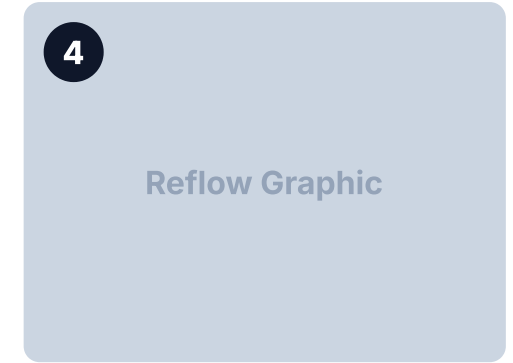
## Solder Paste Stencil

CNC-cut Mylar stencil auto-generated from Gerber file. Fiducial-aligned, paste squeegeed through precision apertures.



## Auto Placement

Components loaded into 3D-printed mould cavities (**₹8 per mould** vs ₹40L PnP machines). Flipped via pins, settled in 3 seconds.



## Reflow Soldering

Controlled ramp to 230°C peak temperature. Solder paste melts forming permanent electrical and mechanical joints.

## Driven by ProtoPulse Suite v1.0.0

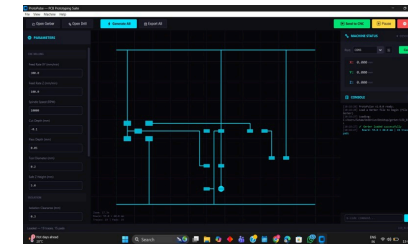
Takes standard Gerber files from any EDA (KiCad, Altium, Eagle) and produces all 4 manufacturing outputs in **under 2 seconds**.

Python 3.12

4,268 Lines of Code

Zero Runtime Deps

Patent-Pending Algorithm



# We win on integration, affordability, and precision.

| CAPABILITY        | BANTAM | LPKF  | BOTFACTORY | VOLTERA | PROTOPULSE |
|-------------------|--------|-------|------------|---------|------------|
| Precision Milling | ✓      | ✓     | --         | --      | ✓          |
| Paste Dispensing  | --     | --    | ✓          | ✓       | ✓          |
| Pick & Place      | --     | --    | ✓          | --      | ✓          |
| Reflow            | --     | --    | --         | ✓       | ✓          |
| Cost / Board      | ₹400+  | ₹400+ | ₹800+      | ₹600+   | ₹150       |
| Turnaround        | Hours  | Hours | Hours      | Hours   | ~60 mins   |

## The Missing Middle

No competitor offers precision milling, paste, placement, and reflow in a single affordable desktop unit. Customers are forced to trade off precision, integration, or cost.

### Bantam & LPKF

Proven milling, but no assembly capabilities. Manual soldering remains.

### BotFactory & Voltera

Conductive ink limits conductivity and resolution. High consumable costs.

# Hardware + Consumables = High LTV.

## 1 Hardware CapEx

Direct sale to universities, R&D labs, and maker spaces. One-time high-value purchase with a target 40-50% gross margin.

## 2 Consumables (Recurring)

V-bits, FR-4 boards, solder paste, Mylar, PLA. High-margin repeat purchases yielding ₹2L-5L per station annually.

## 3 Cloud SaaS & Support

Enterprise tier software features (fleet management, DFM checks), plus annual maintenance contracts and faculty training.

### Unit Economics

**₹8-10L**

EST. COGS

**40-50%**

GROSS MARGIN

**18 mo**

CAC PAYBACK

### Target Audience

**60%**

Colleges & Universities

**30%**

Hardware Startups

**10%**

Prosumers & Makers

# From lab to market.

1

## Months 0-3: MVP & Validation

Finalize CNC isolation milling pivot, release Software v1. Produce 50 sample boards with 90%+ yield target.

2

## Months 3-6: Productization

Develop safety enclosure and dust extraction. Polish UI/UX. File provisional patent for the Gerber-to-Mould algorithm.

3

## Months 6-9: Pilot Deployment

Deploy 5 beta units in partner colleges. Initiate safety certifications. Field-validate unit economics and hardware reliability.

4

## Months 9-12: Market Launch

Establish low-volume assembly line (20 units/month). Secure B2B channel partnerships for university sales.

### THE ASK

# ₹25L

## Seed Round

Funding the first 12 months to build the MVP, deploy 5 pilot units, file IP, and reach a ₹1-1.5Cr Series A milestone.